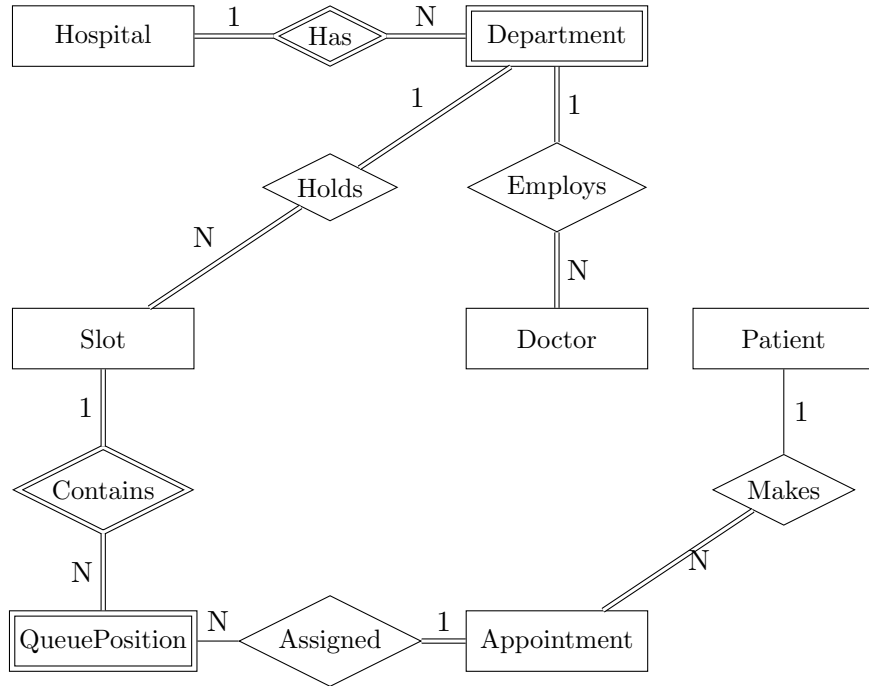


CSC5041 Assignment 1

Concise answers

Question 1 (5 marks)

ER diagram: relationships



Double rectangles/diamonds: weak entities/identifying relationships.

Double lines: total participation. Attributes are shown on the next two pages.

Assignment constraints

Let $\text{slot}(q)$ be the slot owning queue position q , and $\text{Assigned}(a, q)$ mean that q is allocated to appointment a .

$$\boxed{\text{Included}(a, s) \iff \exists q [\text{Assigned}(a, q) \wedge \text{slot}(q) = s]}$$

$$\boxed{\text{Assigned}(a, q_1) \wedge \text{Assigned}(a, q_2) \wedge \text{slot}(q_1) = \text{slot}(q_2) \Rightarrow q_1 = q_2}$$

Thus each included slot has exactly one allocated position; each position is allocated to at most one appointment (the 1 on the Appointment side).

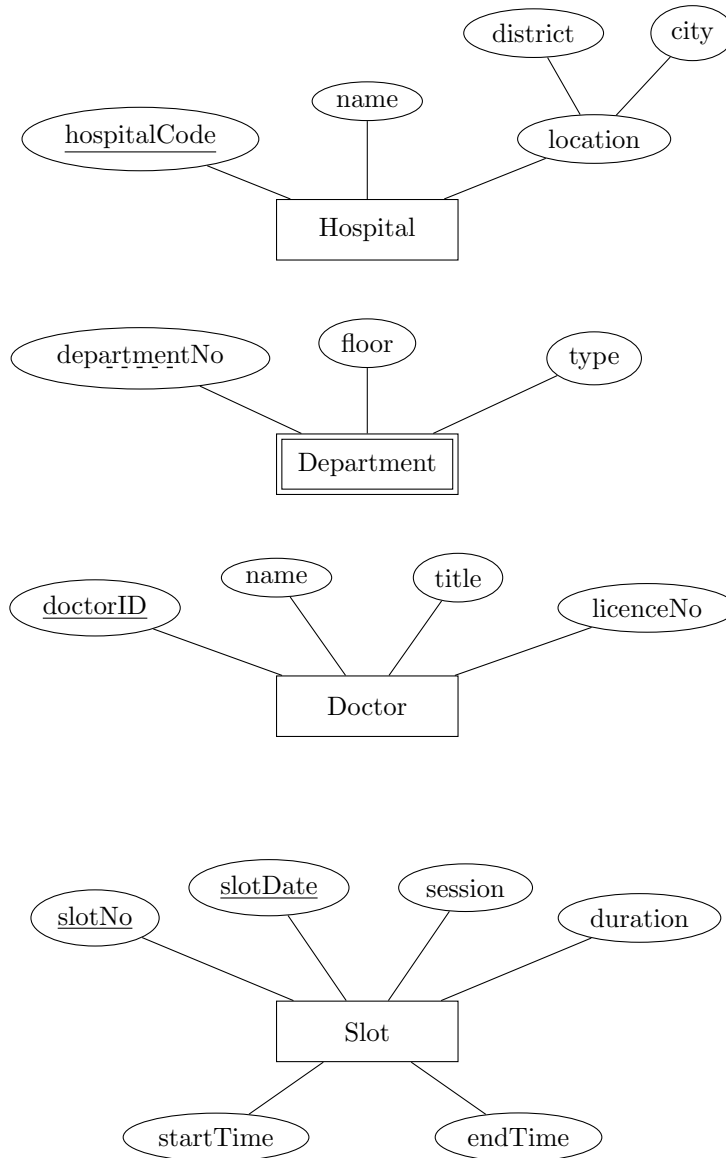
Assumptions: Each appointment contains at least one slot. A slot may have no appointments. Tags are optional. Duration and totalSlots are stored, as requested:

$$\text{duration} = \text{endTime} - \text{startTime}, \quad \text{totalSlots}(a) = |\{s : \text{Included}(a, s)\}|.$$

Slots start and end on the same date; age is derived from dateOfBirth.

Question 1: attributes (1/2)

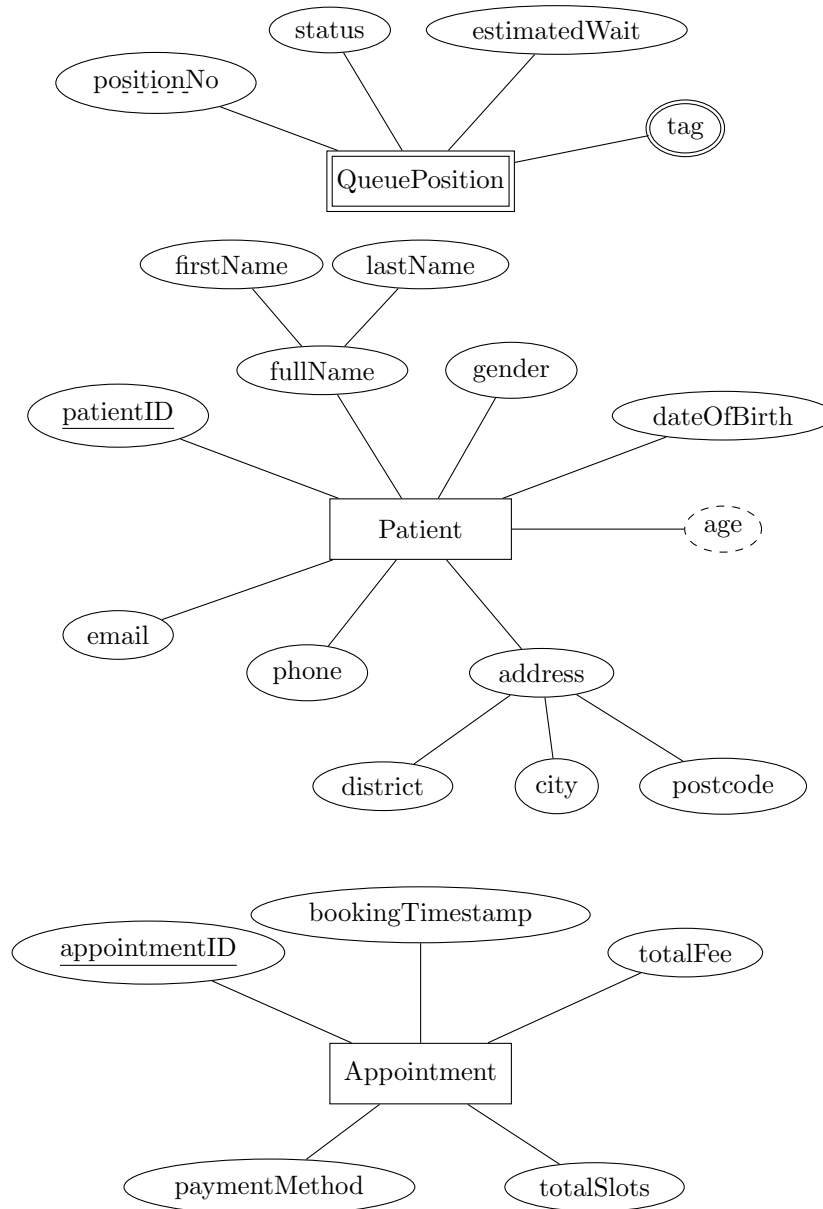
Underlined: key. Dashed underline: partial key. Dashed oval: derived attribute.



$K_{\text{Hospital}} = (\text{hospitalCode}),$
 $K_{\text{Department}} = (\text{hospitalCode}, \text{departmentNo}),$
 $K_{\text{Doctor}} = (\text{doctorID}),$
 $K_{\text{Slot}} = (\text{slotNo}, \text{slotDate}).$
 $\text{session} \in \{\text{morning}, \text{afternoon}\}.$

Question 1: attributes (2/2)

Double oval: multivalued attribute. Repeated entity names denote the same entities.



$$K_{\text{QueuePosition}} = (\text{slotNo}, \text{slotDate}, \text{positionNo}).$$

estimatedWait is measured in minutes; tag may have zero or more values.

Question 2 (4 marks)

Underline = primary key. Composite attributes are flattened; multivalued attributes form a separate relation; derived attributes are omitted.

$E1 (\underline{A1}, A2)$
$E2 (\underline{A9}, \underline{A3}, A4)$
$E3 (\underline{A5})$
$E3_A6 (\underline{A5}, \underline{A6})$
$E4 (\underline{A7}, A8)$
$E5 (\underline{A9}, A12)$
$E6 (\underline{A12}, A14, A15)$
$R1 (\underline{A1}, \underline{A9}, \underline{A3}, A7)$
$R3 (\underline{A5}, \underline{A12}, A16)$

Foreign keys

$$\begin{aligned} E2.A9 &\longrightarrow E5.A9, \\ E3_A6.A5 &\longrightarrow E3.A5, \\ E5.A12 &\longrightarrow E6.A12, \\ R1.A1 &\longrightarrow E1.A1, \\ R1.(A9, A3) &\longrightarrow E2.(A9, A3), \\ R1.A7 &\longrightarrow E4.A7, \\ R3.A5 &\longrightarrow E3.A5, \\ R3.A12 &\longrightarrow E6.A12. \end{aligned}$$

Constraints and mapping

$$\begin{aligned} &\text{UNIQUE}(E5.A12), \quad E5.A12 \neq \text{NULL}; \\ &\pi_{A1}(R1) = \pi_{A1}(E1), \quad \pi_{A7}(R1) = \pi_{A7}(E4); \\ &\pi_{A12}(R3) = \pi_{A12}(E6). \end{aligned}$$

- $R2$: identifying $1 : N$ relationship, absorbed into $E2$ through $A9$.
- $R4$: $1 : 1$, with total participation of $E5$; absorbed into $E5$ through $A12$.
- $R1$: $(A1, A9, A3) \rightarrow A7$ (the 1 on the $E4$ side).
- $A10, A11$ are derived; $A13$ is replaced by $A14, A15$.

Question 3 (6 marks)

All $\bowtie$ without a condition are natural joins. Aggregation: $\gamma_{G; \text{FUNC}(A)}$ groups by G .

1) Car trips in June 2025

$$\pi_{\text{riderName}} \left(\text{Rider} \bowtie \sigma_{\substack{\text{category}='car' \wedge \\ \text{tripDate} \geq '2025-06-01' \wedge \\ \text{tripDate} < '2025-07-01'}} (\text{Trip} \bowtie \text{VehicleType}) \right)$$

2) Maximum payment: only trips with fare ≤ 10

$$G \leftarrow \pi_{\text{riderID}}(\text{Trip}) - \pi_{\text{riderID}}(\sigma_{\text{fare} > 10}(\text{Trip})),$$

$$\boxed{\text{Answer}} \leftarrow \gamma_{\text{riderID}; \text{MAX}(\text{amount})}(G \bowtie \text{Payment}).$$

Here “has made only trips” requires at least one trip; MAX is returned for riders with recorded payments.

3) Riders using every available category

$$C \leftarrow \pi_{\text{category}}(\text{VehicleType}),$$

$$U \leftarrow \pi_{\text{riderID}, \text{category}}(\text{Trip} \bowtie \text{VehicleType}),$$

$$G \leftarrow U \div C,$$

$$\boxed{\text{Answer}} \leftarrow \pi_{\text{riderID}, \text{membershipTier}}(\text{Rider} \bowtie G).$$

Assume the system has at least one vehicle category. Retain riderID to identify each rider.

4) Average age, counting every qualifying trip

$$S \leftarrow \rho_{(\text{startStationID}, \text{zoneID}, \text{capacity}, \text{stationType})}(\text{Station}),$$

$$T \leftarrow \sigma_{\substack{\text{zoneName}='Xuhui' \wedge \\ \text{tripDate} \geq '2025-10-01' \wedge \\ \text{tripDate} \leq '2025-11-02'}} (\text{Trip} \bowtie S \bowtie \text{Zone} \bowtie \text{Rider}),$$

$$V \leftarrow \pi_{\text{tripID}, \text{age}}(T),$$

$$\boxed{\text{Answer}} \leftarrow \gamma_{\text{AVG}(\text{age})}(V).$$

Retaining tripID gives one tuple per trip, even when ages are equal.